

The Application of
Array Processing in Earth Resources

Adrian Lincoln
Fast Firmware Techniques Limited
2 Venture Road, Chilworth Research Centre,
SOUTHAMPTON, Hants, UK SO1 7NP

Ian Curington
Array Productions Ltd
6 Erskin Road, Primrose Hill
LONDON NW3 3HA, UK

Introduction

Array processors are achieving high performance for special scientific programs requiring a large amount of floating point computations. Using techniques such as parallelism, pipelining, microcode control, and VLSI components, high performances are obtained through architectural features rather than in the use of exotic and expensive technology.

The use of array processors within the VAX/VMS environment is discussed and using examples from earth resource analysis some of the parameters and factors associated with obtaining maximum benefit are considered.

This talk is broken down into three sections.

The first part will cover some general aspect of array processors, what they are, why we use them, and how they appear and perform within the VAX environment.

In the second section typical usage, and performance requirements will be outlined by examples from the areas of

- synthetic aperture radar
- image rectification
- resource image data

The final part of the talk will summarise some of the general characteristics of the applications and the factors which feature in the choice of an array processor as a solution for computational intensive processing.

1 Attached Array Processors

What are they?

Why do we need them?

What can they do?

How do they appear to the user?

What is an Attached Array Processor?

An array processor is a computing engine that has been specifically designed to support array type problems. Unlike general purpose computers such as VAX which are required to perform both computing and input/output functions, the array processor need only perform a computing function and therefore can adopt an architecture which is more suited to that type of work. In particular, by reducing the variety of operations that need to be performed, its reduced functional requirement can lead to significant improvements in efficiency through careful design.

One aspect of this restricted capability is the need to provide a host computer which is of a more general purpose nature. This host may be an integral and imbedded component of the array processor or more typically a traditional minicomputer such as the VAX or MicrdVAX. The function of this host computer is primarily to supervise the operations of the attached processor. It will

offload work to the attached array processor in the same way that a manager will offload his typing to a secretary who can perform this function more quickly and efficiently than the manager. Following on from this analogy the manager is left to carry out his own activities for which he is more suited. In a similar manner the host computer is able to continue with other work until it is interrupted by its slave. The host's function will include:

- 1) The transfer of data to and results from the attached processor.
- 2) The control and synchronisation of the attached processor operations.

In terms of the user who may want to take advantage of one of these devices connected to his computer it must be said that in most cases the operation is not transparent. The user must be aware that he is taking advantage not only of the host processor but of the additional slave processor. In this way attached array processors differ from the coprocessor type of devices in many micro and minicomputers.

In a co-processing environment there is a much closer relationship between the host computer and an arithmetic coprocessor. If certain instructions are executed by the host computer these will automatically be passed to the integral coprocessors while the main CPU waits for the result. In the case of an attached array processor no intimate knowledge or interaction of the hardware takes place and work must be given to it specifically. This approach has many advantages in system design, in particular the user can partition a task over several processors which will operate concurrently.* In brief, array processors should be given work that it does best ie floating point computation. Whilst the host should do the general purpose processing, the work for which it was designed.

Why Do we Need Them?

Array processors appeared in the early 1970s when there was a need to increase the performance of floating point computation in areas such as digital signal processing. *Although it was possible to get general purpose computers with the performance required that solution was prohibitively expensive in terms of the overall application requirement and thus uneconomical. It is in these cases of computationally intensive processing where the attached processor can become very viable and an economical solution.

Typical application areas involve either large volumes of data which need to be processed as quickly as possible or small volumes which need to be performed at high speed possibly in real-time.

What Can They Do?

Following on from the previous sections, since there is no requirement for an attached processor to be able to perform the operations associated with general purpose processing they can be designed specifically to perform arithmetic operations in an efficient manner. By adopting architectures which are particularly orientated to floating point computation they can perform this work giving high performance at lower cost. Many of these architectures include separate data and program memories as well as a modest degree of parallelism.

How do they appear to the user?

An attached array processor is a device that is connected to a host computer in much the same way as a normal peripheral device such as a disk or tape drive. This connection can vary in that it may vary the degree of coupling between the two machines.

A loosely coupled processor is invariably a device to which data and work is passed from the host and the results are returned upon completion. A closely coupled device implies a tighter bond between the two processors and to some extent results in the ability to share resources. This in itself can be an advantage or a disadvantage depending upon the mechanisms used to achieve the coupling.

In all cases the attached processor is a slave processor and in terms of operation, resides outside the scope of the normal operating environment as supplied by the manufacturer. As a consequence of this remoteness the operation of the attached processor is not transparent and the user must become involved with the operation of two devices. This operation may sound difficult but in practice is usually quite straightforward.

For example, if the user wishes to run the computationally intensive program he will partition the program (which for well designed structured programs should be trivial) and produce 2 program source files. The computationally intensive program will be compiled so that it could be run on the attached processor whilst the remainder of the application program would be compiled to run on the normal host. The only change to the program will be the inclusion of some subroutine calls in the host program to control the operation of attached processor. As mentioned briefly before the distribution of work between systems can be of enormous benefit for the overall performance of the system.

2 Applications

To provide an indication of how array processors are used, I will outline some typical applications from the area of earth resources which take advantage of these systems. The applications considered below are synthetic aperture radar, image rectification, and resource image data processing.

Before considering these examples in detail there are some general characteristics about these applications that are worth considering.

- 1) Most algorithms use 32-bit floating point format. Although it would be possible and possibly more efficient to use integer formats in some applications, this can often lead to problems in terms of producing portable and therefore easy to use code.
- 2) The input data usually has a precision of 8 to 20 bits.
- 3) The processing which contains a high proportion of FFT and convolution kernels usually requires a precision of 24 to 32 bit. Some of the factors which need to be considered in this processing are signal noise and numerical stability.
- 4) In all these applications there is a vast amount of data to be analysed.

2.1 Synthetic Aperture Radar

Synthetic aperture radar (SAR) is a technique repeatedly for taking one dimensional images from a moving satellite to generate a two dimensional image. The information contained within this image is application dependent and the following discussion outlines some of the general characteristics associated with the processing involved.

Figure 1 illustrates a satellite which is in orbit and is using radar to obtain information about the earth's surface beneath. To be able to build up an image it emits a pulse and detects the echos, repeating this process as it travels in its orbit. The responses from each output pulse are used to generate part of a 2-D image of the observed object beneath.

In the measurement technique and subsequent processing there are 3 factors that have considerable effect on the data that is available and its processing.

- 1) The radar system itself is a broad beam radar (using a small antenna) and receives data in a complex form (amplitude and phase are available). Because of a physical restraint on the output power available and the difficulty with determining the time return of a very narrow pulse the radar system will also use a chirp pulse technique. This chirp pulse consists of an output which varies in frequency over time.

The deconvolution of the return pulse with the ideal output ie a cross correlation will produce a spike given the time at which the return was received the resolution and accuracy of this time is determined to a large extent by the processing used rather than the physics of the system. An obvious advantage of the chirp technique is the ability to output large amounts of energy.

- 2) The satellite is moving, not only is the satellite in orbit but it is also wobbling and rotating. So having emitted a pulse, reflections will be received at various new positions. Another aspect of broad beam systems is that when receiving echos there is no guarantee that they are due to the current pulse. The situation can lead to a multiplicity of data in the input stream.
- 3) There are various geometric factors such as the curvature of the earth and the fact that we may be taking measurements at weird angles which will particularly distort the data received.

Since we are using this as an example of array processing rather than a discussion of SAR, I will briefly outline some of the parameters associated with these types of applications, consider the processing requirements, and indicate types of solution.

Pulse repetition rate: 1700 HZ

For each pulse 5700 samples = 9.69 Mvalues/sec

Each image consist of 2048 records which are typically 8192 data points (complex data). This data is 5 real bits plus 5 imaginary bits equals 10-bits/sample (approximately 100 megabits/sec).

Typical processing associated will include formatting/correction of data, filtering, correction of effects such as range mitigation (curvature of earth), position, multiplicity of data. The output of this processing will be a "raw picture" which will require further processing to extract the information required for the application. Before discussing some of the techniques associated with post processing of the raw picture, it is worth considering some of the characteristics of a typical application. Typically most of the processing consists of 75-80% of FFT's which for a typical image would require 10 gigaflops if the processing was to be done in real-time. For non-real-time work using a 12 megaflop machine, each image requires approximately 40 minutes processing.

SAR is only one detection technique and there are others available such as infrared and invisible light. All of these techniques produce an image which will have different attributes and which serve the different application requirements. For example, these applications may be geological, the search for mineral deposits, weather, and agricultural assessment. One aspect of all these techniques is that we will get a picture from which we need to extract the information required. However, if the image has been obtained from a satellite, it will be distorted by such factors as motion and the curvature of the earth and therefore before starting the processing to extract the information you must present the image in a standard geographical form. These images will need to be realigned to "ground truth" so that the information is presented in a meaningful manner. This process of image rectification is discussed in the next section. «

2.2 Image Rectification

The process of image rectification involves transforming the raw picture image into the new and realigned image. This process involves sampling the raw data to generate this new rectified image. Figure 2 illustrates the type of processing that may be required. The input data may consist of a two dimensional image which is 2K by 8K. To enable us to start the transformation process, it is important that we can identify where our distorted wall picture data was geographically. This process can be helped.

- 1) By knowing where the picture was taken
- 2) By using processing techniques to identify key points.

In essence this process of sampling the raw data to produce a new image involves a "rubber stretching" type technique to remove the distortions of our image.

Much of the work can be seen to involve the sampling of the raw picture and various techniques are available. In the simplest case we could take the nearest value, this could further be refined by using linear interpretation. At the other extreme we could use some kind of convolution technique involving a matrix upto 25 to 25.

The accuracy of our new image to a large extent will be determined by the amount of processing that is carried out in its generation.

2.3 Processing of Resource Image Data

In the previous sections we have discussed SAR techniques in detail and mentioned some of the other techniques which are available. Depending on the information that is required by the user there are various different types of processing that will also be required to extract information from our now rectified image. In this section I will outline two of the types of technique that are used in this post processing. The first is feature extraction where we are looking for geometric factors. Some of these applications may be associated with identifying:

- geological features
- types of reflection
- crops, roads, cities
- shapes - airports.

The techniques associated with this type of processing include scaling/thresholding/edge detection/thinning/thickening/convolution/local area histogramming.

An example of the type of processing that will be carried out is say a landsat image of the great west desert in Australia would be a search for features but without the sand dunes. By the use of special filters it will be possible to remove the sand dune features to expose any underlying structure.

The second type of processing involves image classification (Spectral) this type of processing will be required for applications such as agriculture/fields

- wheat, corn, forests
- snow on mountains to estimate volume of water and spring.
- animal life, this type of processing involves looking at the pixel data in the raw image and using this to determine the characteristics of the image.

This data is further classified using a classifier program which inspects data values, places each pixel into a category and repaints the picture in terms of these categories. This technique can be extended to multiple plan operations where categories might involve the identification of red, green, blue, infrared, SAR data.

In all these techniques where the value of a pixel is associated with some attribute of the image it may be necessary to define a reference which is known, to ensure the integrity for the results obtained.

2.4 Other Examples

Satellite imaging is a technique that is ideal for making observations of the large areas of the earth's surface. Another project which is taking advantage of this is the University of London's 2.5-D project. This project involves using 2 satellite pictures (stereo) to generate 3-dimensional pictures of the surface.

Before leaving this area of satellite imaging there is one aspect of the work which cannot be ignored in the development and verification of new techniques such as SAR. When developing techniques it is important that one can test their operation and it is by the use of simulation techniques that a higher level of confidence can be placed on data received. As in the processing of real data the simulation techniques themselves can involve a significant amount of processing and once again in these areas array processors are of significant benefit.

3 General Characteristics and Factors Incorporated in Choosing and Attached Array Processor

In the applications that have been discussed it is apparent that the processing required is not only related to the computation but also the large data sets involved. The order of processing involved for each element would typically be in the range 5 to 50 floating point operations. Therefore the larger the data sets the larger the computational task.

The philosophy of identifying the processing required and using a machine to which the work can be off-load has been an essential feature of computing in the areas discussed. It is a consequence of the computational requirement that many of these applications which can broadly be described as computationally intensive take advantage of devices such as array processors, acting as computing engines to provide a very cost effective way of providing the processing power needed.

The identification of computationally intensive applications and subsequent use of array processor relies on the ability to structure the program so that those intensive parts can readily be off-loaded from the host.

Some of the factors which have to be considered in the selection of any particular drive can be summarised as follows:

3.1 Raw Arithmetic Processing Power

Any array processor will have a peak arithmetic processing power. It is a fact that depending on the architecture of the machine - many of which are tailored specifically for vector or DSP type problems - the true processing power will vary. Furthermore, the variation of achievable performance can be large. For example, machines which can achieve a high Megaflop rating for FFT routines may achieve as low as 5-10% when running a FORTRAN compiled program.

3.2 Data Set Size and Memory

The relationship between the parameter which define memory such as size, memory access time, bus bandwidth and overall performance can be more significant than the overall arithmetic processing power of the machine. Apart from a small number of applications, many array processors fail mainly due to restrictions imposed by memory. In the applications described in this paper it is obvious that a large amount of data is handled and the efficiency and speed with which this can be done will offend the overall system performance.

In particular, if the memory is unable to hold the complete data set there may be significant overhead in transfer data to/from the host computer and its peripherals.

3.3 I/O - Data Transfer

An attached processor is a slave processor which is controlled by the host. It is given work by the host and will often use data provided by the host - passing results back upon completion.

In some applications this may impose too great a restriction on the system and so may AP's support alternative routes for transferring data particularly if connected to external equipment. It must be remembered that data transfer times in some applications far outway the total processing time!

3.4 Software

To make the system usable it is essential that appropriate software is made available. This may simply be a subroutine library or may be a full FORTRAN development environment.

Although the concept of using a slave processor to provide multi-process support may appear cumbersome and fraught with hidden dangers, systems are now available which provide a straightforward method of using and programmin AP's, so much so that they can be viewed as a natural extension of the host computer and may naturally fall within the structure of the users' program.

Summary-

This paper has provided a brief outline of attached array processors describing how they appear on a typical computer system. By using some examples from the field of earth sources analysis the justification for such a device is made clear both in terms of economic and functionality.

It is clear from the applications that the type of problems for which they are suited will be limited, but due to advances both hardware/software techniques the attached array processor is becoming more a solve processing engine that can now be use! in a much wider range of applications scientific/engineering, industrial, and business.

Satellite in Orbit

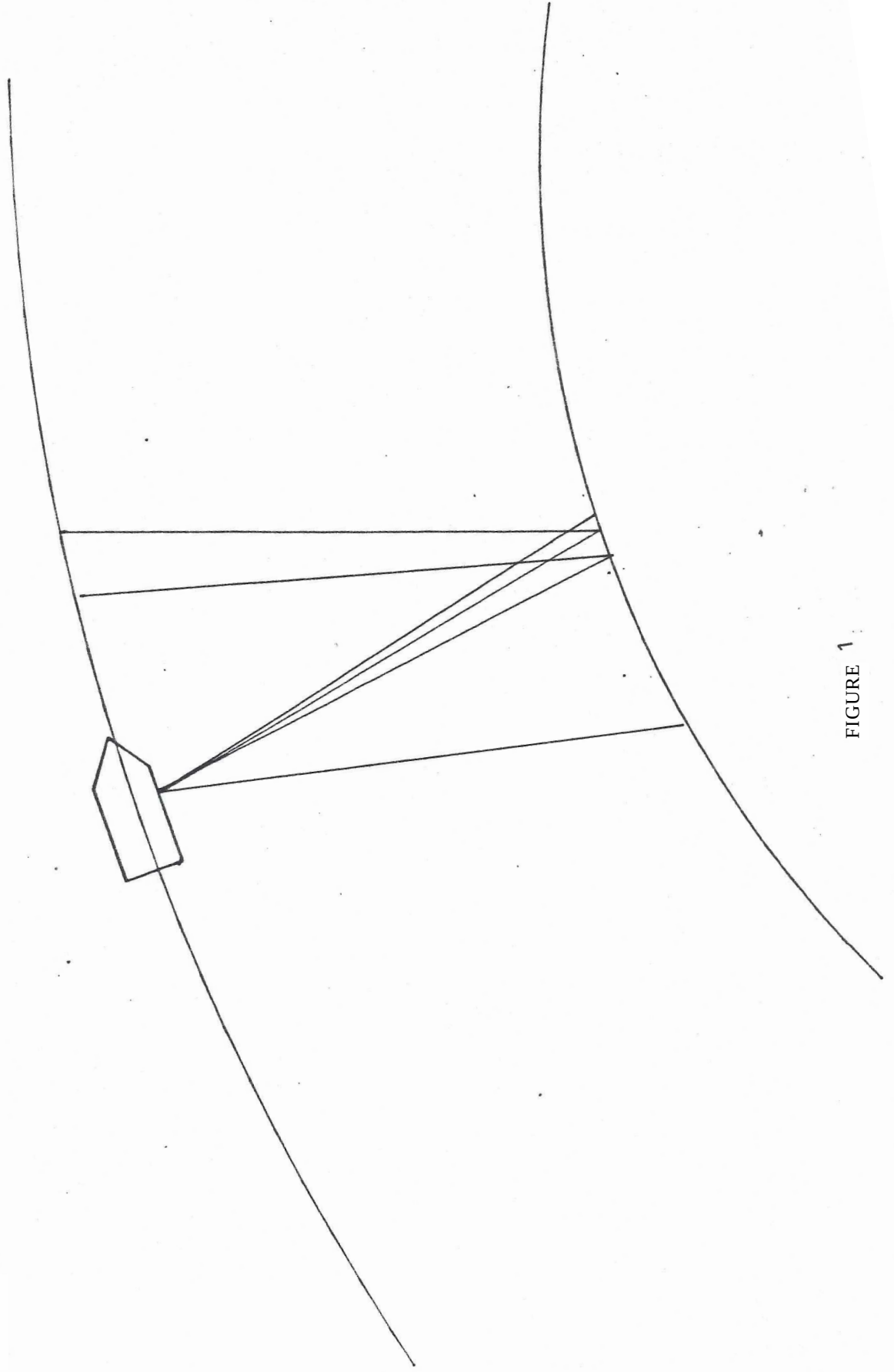
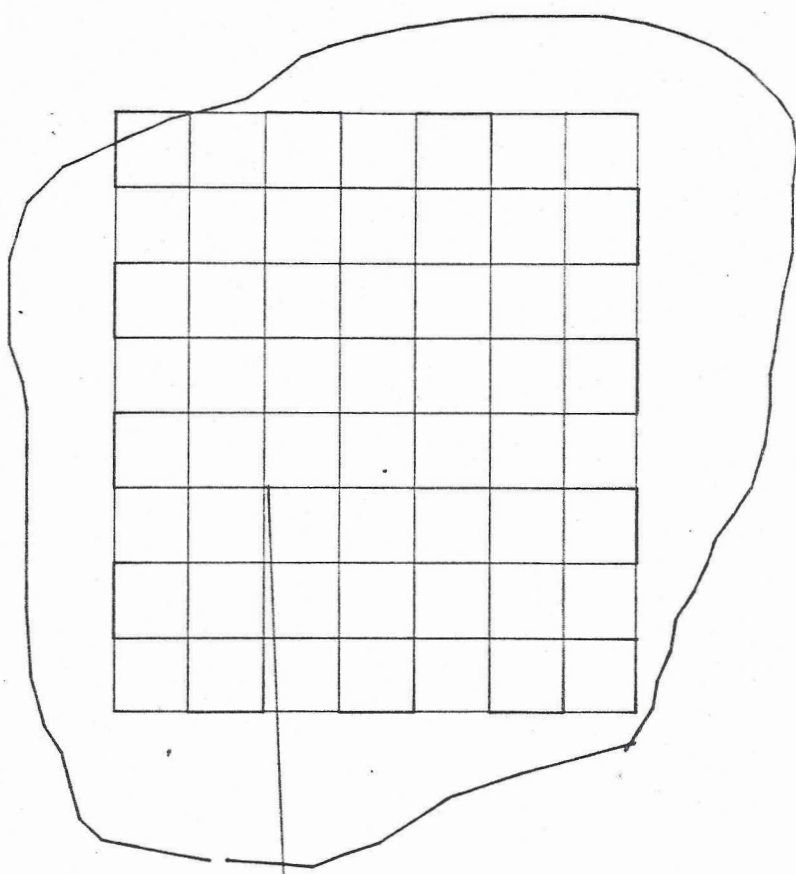
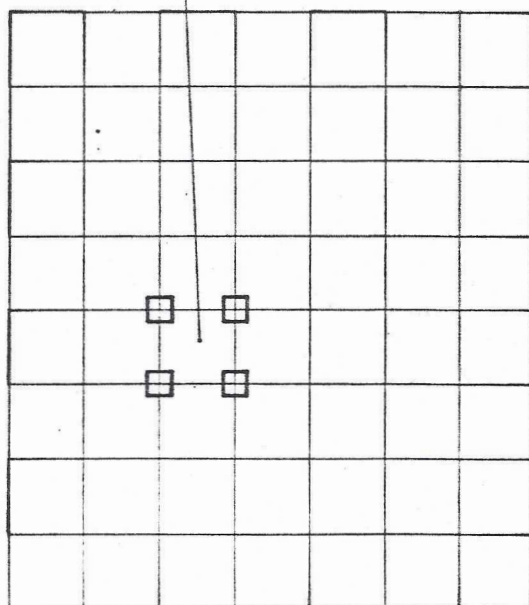


FIGURE 1



Ground Truth Image.



Raw Image

FIGURE 2